

TON OPEN FOOTBALL (\$TOF)

GAME DESIGN & TECHNICAL SPECIFICATION (MVP)

1. Product Overview

Game Type: Competitive Football Manager (Simulation PvP)

Platform: Telegram Mini App

Blockchain: TON

Primary Goal (MVP): Deliver a fast, competitive PvP football simulation with token utility and progression system.

2. Core Game Loop

1. User opens Telegram Mini App
 2. Auto-auth via Telegram
 3. Connect TON wallet
 4. Receives starter squad
 5. Plays PvP match
 6. Wins/loses → earns rewards
 7. Upgrades team
 8. Climbs leaderboard
 9. Repeat (Target loop: 3–5 minutes)
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3. User System

Authentication:

- Telegram ID as primary user ID
- TON wallet address linked

Backend stores: telegram_id, wallet_address, username, elo_rating, coins, tof_balance

4. Team System

Squad Structure:

- 11 Starting players
- 3 Bench players
- Formations: 4-4-2, 4-3-3, 3-5-2

Player Object:

- id, owner_id, position, rarity
- attack, defense, speed, stamina
- overall_rating, level, max_level

Overall Rating Formula:

$(\text{attack} \times 0.35) + (\text{defense} \times 0.30) + (\text{speed} \times 0.20) + (\text{stamina} \times 0.15)$

5. Match Engine (Server-Side Only)

Team Strength =

$(\text{Avg Attack} \times 0.35) +$
 $(\text{Avg Defense} \times 0.30) +$
 $(\text{Formation Bonus} \times 0.15) +$
 $(\text{Avg Stamina} \times 0.10) +$
(Random Factor 0–8%)

Win Probability:

$\text{Team A Strength} / (\text{Team A} + \text{Team B})$

Includes: Winner, Scoreline, Match Report

ELO System: Start 1000, scaled gain/loss, weekly reset.

6. Reward System

Win: Coins + small \$TOF + XP
Loss: Small coins + XP
Draw: Moderate coins + XP
XP unlocks progression tiers.

7. Upgrade System

Upgrade requires Coins + \$TOF
Effect: +1 random stat (within rarity cap)

Rarity Caps:
Common 10
Rare 20
Epic 30
Legendary 40

Burn Logic:
20% of \$TOF burned, 80% to treasury.

8. Pack System

Standard (Telegram Stars)
Premium (Toncoin)
Elite (\$TOF)

Example Probabilities:
Standard: 70% Common, 25% Rare, 5% Epic
Premium: 50% Rare, 35% Epic, 15% Legendary
Elite: 30% Epic, 70% Legendary

9. Economy Structure

Currencies:

- Coins (off-chain)
- Toncoin (on-chain)
- Telegram Stars
- \$TOF (Jetton)

\$TOF Utility:

Upgrades, Tournament Entry, Elite Packs, Marketplace Fees, Battle Pass.

10. Smart Contracts (TON)

- \$TOF Jetton Contract
- Tournament Escrow
- Burn Mechanism
- Treasury Wallet

NFT Player Minting: Phase 2

11. Backend Requirements

Suggested Stack:

- Node.js
- PostgreSQL
- Redis
- TON SDK

Core Modules:

Auth, Match Engine, ELO System, Reward Distribution,
Pack Probability Engine, Upgrade Logic, Anti-cheat Validation.

12. Anti-Cheat

- Server-side match calculations
 - Match cooldown timers
 - Wallet-based identity
 - No client-trusted stats
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13. MVP Scope

Included:

Telegram login, Wallet connect, Starter team, PvP matchmaking, Match simulation, Upgrades, Pack purchase, Leaderboard.

Excluded:

NFT minting, Marketplace, Clan system, Live animation.

14. Development Timeline

Week 1–2: Architecture + DB + Wallet

Week 3–5: Player system + Match engine + ELO

Week 6–7: Upgrade + Packs

Week 8: Testing + Balancing

Estimated Total: 8 Weeks
